

AmirPouya Moeini

Ph.D. student in Engineering @ University of Cambridge

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Research Interests

I am broadly interested in **high-dimensional probability**, **information theory**, and **combinatorics**. My PhD studies error-exponent limits in information theory (achievability and converse), using tools from probabilistic combinatorics, with a particular focus on mismatched and universal settings.

Education

Ph.D. in Engineering

📅 Oct 2023 – Ongoing

🏛️ University of Cambridge, UK — Probabilistic Systems, Information, and Inference Group (PSI²)

Supervisor: Prof. Albert Guillén i Fàbregas

B.Sc. in Computer Engineering (Minor in Applied Mathematics)

📅 Sep 2018 – Jun 2023

🏛️ Sharif University of Technology, Iran

Thesis: *Information-theoretic bounds for online learning* (20/20)

Publications

“Two-Class Joint Source-Channel Coding: Expurgated Exponents with i.i.d. Distributions” [🔗 \[Link\]](#)

Moeini, A. and Guillén i Fàbregas, A., 2026 International Zurich Seminar on Information and Communication (IZS).

“Dual-Domain Exponent of Maximum Mutual Information Decoding” [🔗 \[Link\]](#)

Moeini, A. and Guillén i Fàbregas, A., 2025 IEEE Information Theory Workshop (ITW).

“Class-Based Expurgation Attains Csiszár’s Expurgated Source-Channel Exponent” [🔗 \[Link\]](#)

Moeini, A. and Guillén i Fàbregas, A., 2025 IEEE Information Theory Workshop (ITW).

Academic Visits

🏛️ National University of Singapore, Singapore

October 2025

School of Computing – Institute of Data Science, visiting Dr. Jonathan Scarlett

🏛️ University of Brescia, Brescia, Italy

September 2025

Department of Information Engineering, visiting Dr. Marco Dalai

Honours and Awards

2023	Cambridge International Trust – Full Scholarship ($\approx$£62,000 per year)	Cambridge, UK
2022	Summer ASci @ Aalto University Fellowship	Espoo, Finland
2018	Gold Medal <i>12th International Olympiad on Astronomy and Astrophysics (IOAA)</i>	Beijing, China
2017	Gold Medal <i>13th National Astronomy and Astrophysics Olympiad</i>	Iran

Selected Courses

🏛️ University of Cambridge

Computational statistics and machine learning, Advanced information theory and coding

🏛️ Sharif University of Technology

Probability

Information theory and coding, Information-theoretic methods in high-dimensional statistics, High-dimensional probability, Graphical models, Stochastic processes, Bayesian methods in statistics

Mathematics

Real analysis, Algebra, Linear algebra, Applied linear algebra, Probability & statistics

Machine Learning

Machine learning theory, Artificial intelligence, Computer simulation, Modern information retrieval

Computer Science

Design algorithms, Data structures and algorithms, Compiler design, Database Design

🏛️ Audited Courses

Probabilistic combinatorics, Entropy methods in combinatorics, Ramsey theory, Random discrete structures (*Cambridge*)

Teaching Experience

Supervision  University of Cambridge

- Information Theory and Coding (3F7), Michaelmas 24, 25
- Data Transmission (3F4), Lent 25, 26

Teaching Assistant  Sharif University of Technology

- Information theory & Coding, Fall 22
- Theoretical Machine learning, Fall 22
- Statistical Machine learning, Spring 22
- Mathematical Statistics, Fall 21
- Data Structure & Algorithm, Spring 21
- Linear algebra, Spring 21
- Regression Analysis, Spring 21
- Artificial intelligence, Fall 21, Spring 22

Research Experience

Approximate Bayesian Computation in the Presence of Outliers

 Jul 2022 – Oct 2022

- Probabilistic Machine Learning Group, Aalto University, Finland

- Supervisor: *Prof. Samuel Kaski*.

- Developing computationally efficient algorithms that enhance the robustness of Approximate Bayesian Computation (ABC) inference under outliers and model misspecification.

Information-Theoretic Bounds for Online Learning Problems

 Oct 2021 – Mar 2023

- Machine Learning Laboratory, Sharif University of Technology, Iran

- Supervisors: *Dr. Mahdieh Soleymani* and *Dr. Hassan Hafez-Kolahi*.

- We introduce information-theoretic tools for online learning with adversarial data and derive performance upper bounds via conditional mutual information.

Skills

Programming Language: Python (*advanced*), Java (*advanced*), C / C++, SQL.

Statistical Programming Languages: R (*advanced*), Julia (*advanced*), Stata, MatLab & Octave.

Tools and Software: MS Excel, L^AT_EX, Jupyter NB / G Colab, PyTorch, Git / GitHub

Language: Farsi/Persian (*native*), English (*fluent*), French (*basic*).